

Riley Brown

rydb@proton.me · rbrown71@nebraska.edu · ry3b.github.io · github.com/ry3b

Education

University of Nebraska-Lincoln

Aug 2023–May 2027
(expected)
Lincoln, NE

B.S. Mathematics and Data Science, Minor in Computer Science

- GPA 3.92/4.00, University Honors Program

Budapest Semesters in Mathematics

Jan 2026–Jun 2026
Budapest, Hungary

Study abroad, mathematics

Research Experience

Undergraduate Researcher - Quantum LDPC Codes

Fall 2026

University of Nebraska-Lincoln advisor: *Professor Christine Kelley*

Lincoln, NE

- Testing whether collision-free geodesic-flow operators on Ramanujan complexes can sharpen iterative-decoder thresholds for quantum LDPC codes

Research Assistant — pseudorandomness

Summer 2026

North Carolina State University advisor: *Professor Chin Ho Lee*

Raleigh, NC

- Studied Ta-Shma’s epsilon-balanced code construction with Prof. Chin Ho Lee as a research partner
- Found simpler constructions and minor improvements to the Hsieh-Mohanty-Zheng construction

Research Assistant, t-uniform hypergraph degree sequences and FPT algorithms

Spring 2026

Budapest Semesters in Mathematics advisor: *Professor Istvan Miklos*

Budapest, Hungary

- Proved t-uniform hypergraphicality is FPT in (k, t) for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via bounded-dimension spectra and balancing hinge-flips

Summer Researcher — algorithm design

Summer 2025

Harvey Mudd College advisor: *Professor Tim Randolph*

Claremont, CA

- Improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics (team of four)
- Sharpened branching strategies and base-case parameters for graph-theoretic reductions

Investigating the Flow of Opinions in a Diagonal Grid Graph Using a Modified Deffuant-Weisbuch Model

Fall 2024

University of Nebraska-Lincoln, Directed Reading Program advisor: *Nick White*

Lincoln, NE

- Extended the project beyond its initial reading goals by implementing a modified Deffuant-Weisbuch model in Python

- Ran systematic parameter variation experiments showing that opinions invariably converge to consensus on diagonal grid graphs
- Proved the discovered properties and identified future directions on alternative graph topologies

Undergraduate Researcher — tiling theory

Summer 2024

University of Nebraska-Lincoln advisor: *Professor Tri Lai*

Lincoln, NE

- Applied Ciucu’s Factorization Theorem to decompose tiling enumeration problems into tractable subproblems, working up through independent study of survey papers
- Used Randall paths to determine hexagon-half parameters based on hole parity along the symmetric axis
- Computed exact tiling counts by applying the Lindstrom-Gessel-Viennot theorem to determinants of binomial coefficient matrices

Preprints and Publications

Fixed-Parameter Tractability of t-Uniform Hypergraphicality

Riley Brown, Istvan Miklos

arXiv preprint, 2026

arXiv

Awards and Honors

D&F Eastman Scholarship

Mathematics Department recognition

Regents Scholarship

Nebraska Career Scholarship

Dean’s List

all semesters

University Honors Program

Talks

Algorithms for the Sumset Recognition Problem

2025

Harvey Mudd College REU Symposium

Combinatorial Game Theory

2025

UNL Directed Reading Program Symposium

Mathematics at UNL

2025

Marshall Recruitment Talk for High School Students

The Modified Deffuant-Weisbuch Model

2024

UNL Directed Reading Program Symposium

Lozenge Tilings of a Hexagon with Holes

2024

UNL Math Club

Teaching

Teaching Assistant, Calculus I <i>University of Nebraska-Lincoln</i>	2025
Tap Dance Instructor <i>Chase the Spirit of Dance, Lincoln NE and Create 308, Grand Island NE (2018-2024)</i>	2024
Teaching Assistant, Calculus II	
Learning Assistant, Analysis	
Teaching Assistant, Honors Calculus II	

Activities

Putnam Exam Practice <i>University of Nebraska-Lincoln</i>	2024
Quiz Bowl Club <i>literature, modern art, mathematics, computer science</i>	2024
Directed Reading Program <i>combinatorial game theory with Noah Walker</i>	2024
Math Club <i>University of Nebraska-Lincoln</i>	2023

Projects

leanverify <i>Verification kernel for an LLM-driven proof loop: checks a Lean 4 proof against Mathlib, rejecting sorries and smuggled axioms</i>	python, lean 4 · 2026
Poker Trainer <i>A terminal and Tkinter drill trainer for poker skills: pot odds, combos, equity, bet sizing, push/fold and ICM, with an adaptive scheduler that weights drills by weakness and staleness.</i>	python · 2026

Selected Coursework

Mathematics. Bijective Partition Theory, Discrete Mathematics, Matroid Theory, Advanced Combinatorics, Conjecture and Proof, Discrete and Convex Geometry, Graph Theory, Group Theory, Probability Theory, Stochastic Processes, Elementary Analysis, Intro to Modern Algebra, Linear Algebra, Differential Equations, Multivariable Calculus, Math in the City, **Algebra I**, **Coding Theory**

Computer Science. **Development and Analysis of Algorithms**, Automata, Computation, and Language, Data Structures and Algorithms, Data Analysis, Math for Machine Learning, Generative AI, Computational Game Theory

Skills

AI for mathematics. Lean 4 and Mathlib verification tooling, LLM-assisted proof search, agentic research harnesses

Languages. Python, MATLAB, Java, SQL, LaTeX